

- 46.** For each of the naturally occurring lactones below, draw the structure of the compound that would result from hydrolysis using aqueous base.

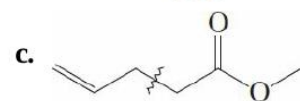
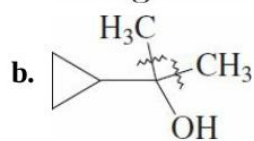
Sedanenolide, major contributor to the flavor of

50. Give the products of the reactions of pentanamide with the reagents given in [Problem 39](#), parts (a, e, f). Repeat for *N,N*-dimethylpentanamide.

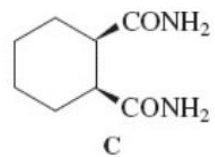
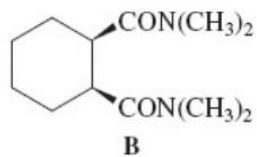
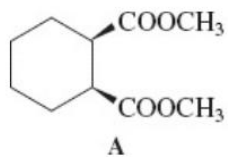
51. Formulate a mechanism for the acid-catalyzed hydrolysis of 3-methylpentanamide shown on [p. 991](#). (**Hint:** Use as a model the mechanism for general acid-catalyzed addition–elimination presented in [Section 19-7](#).)

52. What reagents would be necessary to carry out the following transformations? **(a)** Cyclohexanecarbonyl chloride $\rightarrow$ 1-cyclohexyl-1-pentanone; **(b)** 2-butenedioic (maleic) anhydride $\rightarrow$ (Z)-2-butene-1,4-diol; **(c)** 3-methylbutanoyl bromide $\rightarrow$ 3-methylbutanal; **(d)** benzamide $\rightarrow$ 1-phenylmethanamine; **(e)** propanenitrile $\rightarrow$ 3-hexanone; **(f)** methyl propanoate $\rightarrow$ 4-ethyl-4-heptanol.

53. In each of the following retrosynthetic disconnections (wavy), show reactions that would make the indicated C-C-C bonds.



54. Upon treatment with LDA followed by protonation, compounds A and B undergo cis-trans isomerization, but compound C does not. Explain.



57. Show how you might synthesize chlorpheniramine (see also [p. 762](#)), a powerful antihistamine used in several decongestants, from each of carboxylic acids A and B. Use a different carboxylic amide in each synthesis.

- 61.** The removal of the C17 side chain of certain steroids is a critical element in the synthesis of a number of hormones, such as testosterone, from steroids in the relatively readily available pregnane family.

63. Propose a synthetic scheme for the conversion of lactone A into amine